

[5059]-509

B.E. (Civil Engineering)**MATRIX METHODS OF STRUCTURAL ANALYSIS****(2012 Pattern) (Elective - II)***Time : 2½ Hours]**[Max. Marks : 70**Instructions to the candidates:-*

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Solve the following system by Gauss-Jordan Method [6]

$$x + y + z = 5$$

$$2x + 3y + 5z = 8$$

$$4x + 5z = 2$$

- b) Analyse the beam ABC shown in Figure 1 using flexibility matrix method. AB = 3 m and BC = 6 m. Take EI = constant [6]

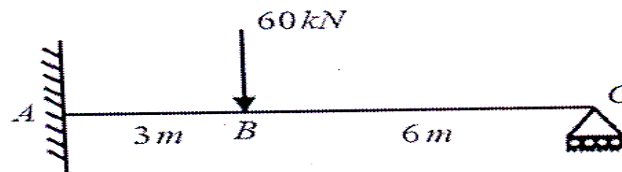


Figure 1

- c) A rod is composed of an aluminum section rigidly attached between steel and bronze as shown in Figure 2. If the cross-section area of rod is 800 mm² determine nodal displacements. Take $E_{st} = 210$ GPa, $E_{Al} = 70$ GPa and $E_{br} = 110$ GPa. [8]

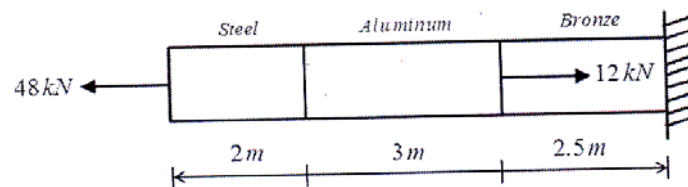


Figure 2

P.T.O.

OR

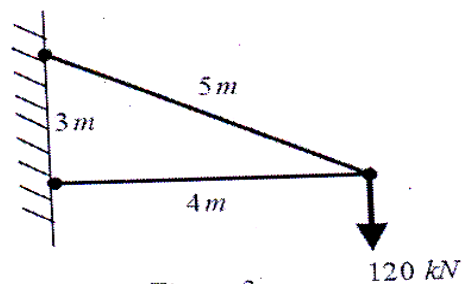
Q2) a) Solve the following system by Gauss-Elimination Method [6]

$$x + y + z = 5$$

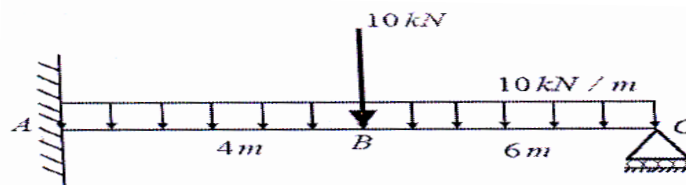
$$2x + 3y + 5z = 8$$

$$4x + 5z = 2$$

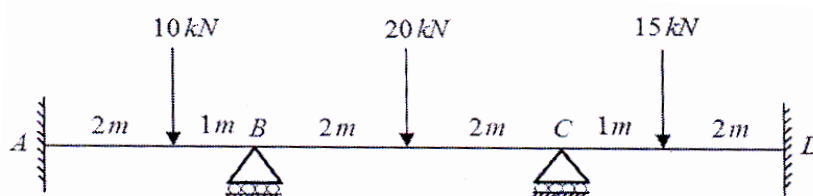
b) Find the vertical and horizontal deflection at point C for the two member truss as shown in Figure 3. Area of inclined member is 2000 mm^2 whereas horizontal member is 1600 mm^2 . Take $E = 200 \text{ GPa}$ [6]



c) Analyse the beam ABC shown in Figure 4 using flexibility matrix method. Take $EI = \text{constant}$. [8]

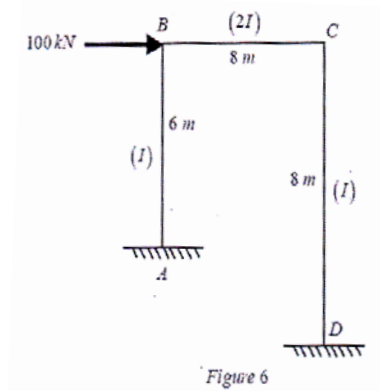


Q3) Analyze the continuous beam ABCD as shown in Figure 5 using stiffness matrix method. Take EI constant. Draw BMD [18]



OR

- Q4)** Determine the unknown joint displacements of the portal frame as shown in Figure 6 using stiffness matrix method. Take EI constant. [18]



- Q5)** Derive the stiffness matrix and transformation matrix of two noded grid element of with 06 D.O.F., length L , flexural rigidity EI and torsional rigidity GJ . [16]

OR

- Q6)** Analyze the grid structure ABC as shown in Figure 7 using stiffness matrix method. Take $EI = 2 \times 10^5 \text{ kN.m}^2$ and $GJ = 1.2 \times 10^5 \text{ kN.m}^2$. [16]

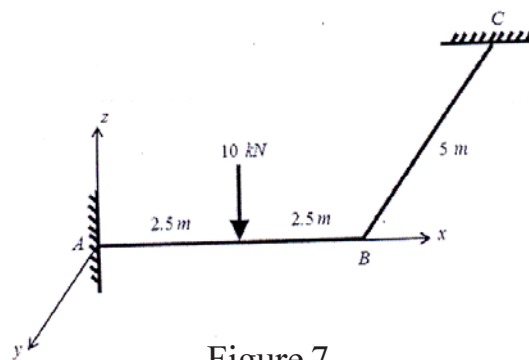


Figure 7

- Q7)** For the truss shown in Figure 7, use stiffness matrix method to determine the deflections at the loaded joint. Take $E = 200 \text{ GPa}$ and c/s area of all members 1000 mm^2 . [16]

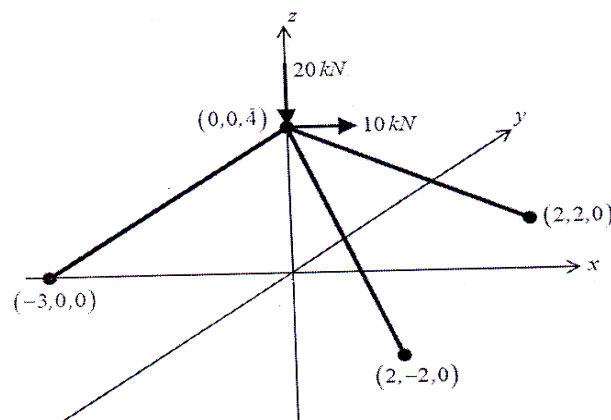


Figure 8

OR

- Q8) a)** A beam of span '8m' is fixed at both ends 'A' and 'B' and supports a uniformly distributed load of 10 kN/m over the entire span. Estimate the deflections of quarter span intervals using second order central difference formula. [8]

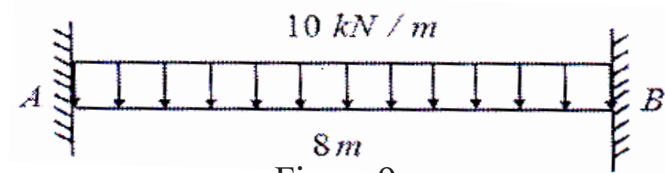


Figure 9

- b)** Estimate the lowest buckling load 'P' of a uniform pin ended column of length 'L = 10 m', cross-sectional area 100×100 mm and $E = 200$ GPa using three sub intervals. Apply finite difference method. [8]

